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Heat dissipation device and electronic apparatus

Abstract

A heat dissipation device includes a body and at least two first capillary structures. The body includes an accommodation chamber. The accommodation chamber includes a heat dissipation medium. The at least two first capillary structures are stripe-shaped and arranged in the accommodation chamber along a flow direction of the heat dissipation medium. A first space is formed between two neighboring first capillary structures. A liquid heat dissipation medium in the two neighboring first capillary structures is evaporated into the first space.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Chinese Patent Application No. 202210114090.1, filed on Jan. 30, 2022, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to the heat dissipation technology field and, more particularly, to a heat dissipation device and an electronic apparatus.

BACKGROUND

(3) A heat dissipation device is a common device. The existing heat dissipation device has poor heat dissipation performance.

SUMMARY

(4) Embodiments of the present disclosure provide a heat dissipation device, including a body and at least two first capillary structures. The body includes an accommodation chamber. The accommodation chamber includes a heat dissipation medium. The at least two first capillary structures are stripe-shaped and arranged in the accommodation chamber along a flow direction of the heat dissipation medium. A first space is formed between two neighboring first capillary structures. A liquid heat dissipation medium in the two neighboring first capillary structures is evaporated into the first space.

(5) Embodiments of the present disclosure provide an electronic apparatus, including a heat dissipation device and a heat generation element. The heat dissipation device includes a body and at least two first capillary structures. The body includes an accommodation chamber. The accommodation chamber includes a heat dissipation medium. The at least two first capillary structures are stripe-shaped and arranged in the accommodation chamber along a flow direction of

the heat dissipation medium. A first space is formed between two neighboring first capillary structures. A liquid heat dissipation medium in the two neighboring first capillary structures is evaporated into the first space. The heat dissipation device is configured to dissipate heat of the heat generation element.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a schematic structural diagram of a cooling device according to embodiments of the present disclosure.
- (2) FIG. 2 illustrates a schematic structural diagram of a cooling device according to embodiments of the present disclosure.
- (3) FIG. 3 illustrates a schematic structural diagram of a cooling device according to embodiments of the present disclosure.
- (4) FIG. 4 illustrates a schematic structural diagram of a cooling device according to embodiments of the present disclosure.

REFERENCE NUMERALS

- (5) TABLE-US-00001 110 Body 111 Accommodation 112 First space chamber 113 Operation wall 114 Second space 115 Third space 120 First capillary structure 121 Recess 130 Second capillary structure

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (6) The technical solution of the present disclosure is further described in detail with reference to the accompanying drawings and embodiments of the present disclosure.
- (7) In embodiments of the present disclosure, unless otherwise specified and limited, the term “connected” should be interpreted broadly. For example, the term “connected” can include an electrical connection, a communication between two elements, a direct connection, or an indirect connection via an intermediate medium. For those of ordinary skilled in the art, the specific meaning of the term can be understood according to a specific situation.
- (8) The term “first\second\third” in embodiments of the present disclosure is only used to distinguish similar objects and does not represent a specific order for the objects. “First\second\third” can exchange a specific order or sequence if allowed. Objects that are distinguished by “first\second\third” can be interchanged under an appropriate situation. Thus, embodiments of the present disclosure described herein can be implemented in an order other than those illustrated or described herein.
- (9) An electronic apparatus of embodiments of the present disclosure can be described in detail below with reference to FIGS. 1 to 4.
- (10) A heat dissipation device includes a body **110** and at least two first capillary structures **120**. The body **110** includes an accommodation chamber **111**. The accommodation chamber **111** can accommodate a heat dissipation medium. The at least two first capillary structures **120** can be strip-shaped. The at least two first capillary structures can be arranged in the accommodation chamber **111** along a flow direction of the heat dissipation medium. A first space **112** is formed between two neighboring first capillary structures **120**. The liquid heat dissipation medium in the two neighboring first capillary structures **120** can be evaporated to the first space **112**. The first space **112** can form a flow channel of the gaseous heat dissipation medium. The liquid heat dissipation medium in the two neighboring first capillary structures **120** can quickly form the gaseous heat dissipation medium in the first space **112**, which can flow. Thus, a route and time for forming the gaseous heat dissipation medium from the liquid heat dissipation medium can be greatly reduced, and the heat dissipation capability of the heat dissipation device can be improved.
- (11) In embodiments of the present disclosure, a structure of the body **110** is not limited. For

example, the body **110** can be a cuboid structure. For another example, the body **110** can be a tube-shaped structure.

(12) In embodiments of the present disclosure, a type of the heat dissipation medium is not limited as long as the heat dissipation medium can be converted between a gaseous state and a liquid state. For example, the heat dissipation medium can be water.

(13) In some embodiments, a flow direction of the heat dissipation medium can be a direction in which the heat dissipation medium flows between an evaporation end of the body **110** and a condensation end of the body **110**. For example, when the body **110** has a tube-shaped structure, the flow direction of the heat dissipation medium can be a length direction of the tube-shaped structure.

(14) In embodiments of the present disclosure, a structure of the first capillary structure **120** is not limited.

(15) For example, the first capillary structure **120** can include a first stripe-shaped member and a second stripe-shaped member. The second stripe-shaped member and the first stripe-shaped member can be arranged alternatively to form the first capillary structure **120**.

(16) In some embodiments, a cross-section of the second stripe-shaped member can be the same as or different from a cross-section of the first stripe-shaped member. For example, the cross-section of the second stripe-shaped member and the cross-section of the first stripe-shaped member can be both circular or elliptical. A diameter of the second stripe-shaped member can be different from a diameter of the first stripe-shaped member.

(17) In some embodiments, the first capillary structure **120** may further include a third stripe-shaped member. The third stripe-shaped member can be staggered with the first stripe-shaped member and the second stripe-shaped member to form the first capillary structure **120**.

(18) A cross-section of the third stripe-shaped member, the cross-section of the second stripe-shaped member, and the cross-section of the first stripe-shaped member can be the same or different. For example, the third stripe-shaped member, the first stripe-shaped member, and the second stripe-shaped member can be column-shaped. At least two of the first stripe-shaped member, the second stripe-shaped member, and the third stripe-shaped member can have different diameters.

(19) In embodiments of the present disclosure, a material of the first capillary structure **120** is not limited. For example, the first capillary structure **120** can have a non-metal structure. Thus, the first capillary structure **120** can have absorbability to retain the liquid heat dissipation medium in the first capillary structure **120**.

(20) In some embodiments, a manner of forming the first space **112** between two neighboring first capillary structures **120** is not limited. For example, as shown in FIG. 1, two ends where two neighboring first capillary structures **120** contact with each other are non-planar. Under the contact of the neighboring two first capillary structures **120**, the first space **112** can be formed by the contacted ends of the two neighboring first capillary structures **120**. For another example, as shown in FIG. 3, a first interval is formed between the two neighboring first capillary structures **120**.

(21) In some embodiments, an arrangement of the at least two first capillary structures **120** is not limited.

(22) For example, the at least two first capillary structures **120** can be arranged in a first direction in parallel. The first direction can satisfy a perpendicular condition with a flow direction of the heat dissipation medium.

(23) In some embodiments, the first direction is not limited. For example, the flow direction of the heat dissipation medium can be a length direction of the body **110**, and the first direction can be a width direction of the body **110**. In some embodiments, the perpendicular condition can be perpendicular or substantially perpendicular.

(24) For another example, the at least two first capillary structures **120** can be arranged in parallel in a second direction. The second direction can satisfy a perpendicular condition with the flow direction of the heat dissipation medium.

(25) In some embodiments, the second direction is not limited. For example, the flow direction of the heat dissipation medium can be the length direction of the body **110**, and the second direction can be a thickness direction of the body **110**. The second direction can satisfy a perpendicular condition with the first direction.

(26) In some embodiments, a cross-section of the first capillary structure **120** in the flow direction of the heat dissipation medium can remain unchanged or can be varied. For example, thicknesses of the at least two first capillary structures **120** in the evaporation region can be greater than thicknesses of the at least two first capillary structures **120** in the condensation region. Widths of the at least two first capillary structures **120** can remain unchanged in the flowing direction of the heat dissipation medium. Thus, more liquid heat dissipation medium can be stored in the evaporation region through the first capillary structure **120** with a larger cross-section, and more liquid heat dissipation medium can absorb heat to evaporate. In the condensation region, the first capillary structure **120** can be mainly configured to transfer the liquid heat dissipation medium to return the liquid heat dissipation medium to the evaporation region. By setting the cross-section of the first capillary structure **120** in the condensation region to be smaller, the space of the gaseous heat dissipation medium in the condensation region can be increased. More gaseous heat dissipation medium can be rapidly condensed in a relatively large space to form the liquid heat dissipation medium. By setting the thicknesses of the at least two first capillary structures **120** in the evaporation region to be greater than the thicknesses of the at least two first capillary structures **120** in the condensation region, the heat dissipation ability of the heat dissipation device can be greatly improved.

(27) In embodiments of the present disclosure, as shown in FIGS. **1** and **3**, the at least two first capillary structures **120** are arranged in parallel on a side of an operation wall **113** of the body **110** in the first direction. The operation wall **113** can be a wall body on a side where the body **110** is connected to a heat generation element. The heat dissipation capability of the heat dissipation device to the heat generation element can be greatly improved through the at least two first capillary structures **120**.

(28) In some embodiments, as shown in FIG. **1** and FIG. **2**, with the same structure of the body **110**, if one first capillary structure **120** is arranged in the accommodation chamber **111**, the width of the first capillary structure **120** is set to be larger in the first direction to cause the heat dissipation device to have a larger heat dissipation capability. Two ends of the first capillary structure **120** can form third spaces **115** with sidewalls of the body **110** in the first direction A. The liquid heat dissipation medium in the two ends of the first capillary structure **120** can evaporate into the third space **115** to form the gaseous heat dissipation medium to flow. If the width of the first capillary structure **120** in the first direction is set too large, a distance between a part of the liquid heat dissipation medium in the first capillary structure **120** and the third space **115** can be too large to be evaporated into the third space **115** to form the gaseous heat dissipation medium to flow, which can greatly lower the heat dissipation capability of the heat dissipation device. Thus, if one first capillary structure **120** is arranged in the accommodation chamber **111**, the width of the first capillary structure **120** in the first direction A can be limited. Even if the width of the first capillary structure **120** in the first direction A is set to be relatively large, the heat dissipation capability of the heat dissipation device can be low because a part of the liquid heat dissipation medium in the first capillary structure **120** cannot evaporate to form the gaseous heat dissipation medium. Meanwhile, the gaseous heat dissipation medium can be condensed on an inner wall of the accommodation chamber **111** to form the liquid heat dissipation medium. The liquid heat dissipation medium can flow back to the first capillary structure **120** along the inner wall of the accommodation chamber **111**. Since the width of the first capillary structure **120** cannot be set to be large in the first direction A, the third space **115** can be relatively large, a route of the liquid heat dissipation medium flowing back to the first capillary structure **120** through the third space **115** along the inner wall of the accommodation chamber **111** can become longer. Thus, the time for the

liquid heat dissipation medium to flow back to the first capillary structure **120** can be greatly increased. Therefore, a period of the liquid heat dissipation medium evaporating again to form the gaseous heat dissipation medium can be increased, which greatly lowers the heat dissipation efficiency of the heat dissipation medium and lowers the heat dissipation ability of the heat dissipation device. As shown in FIG. **1**, the at least two first capillary structures **120** are arranged in the first direction. The first space **112** is formed between the two adjacent first capillary structures **120**. Thus, the liquid heat dissipation medium in the two adjacent first capillary structures **120** can be rapidly evaporated to the first space **112** to form the gaseous heat dissipation medium to flow. Since the at least two first capillary structures **120** are arranged, a width of a single first capillary structure **120** in the first direction can be greatly reduced. Thus, the liquid heat dissipation medium in the first capillary structures **120** can be rapidly converted into the gaseous heat dissipation medium. The widths of the at least two first capillary structures **120** in the first direction can be set to be relatively large. Thus, the route of the liquid heat dissipation medium flowing back to the first capillary structures **120** along the inner wall of the accommodation chamber **111** can be shortened, which greatly reduces the time of the liquid heat dissipation medium flowing back to the first capillary structures **120**. Therefore, the period of the liquid heat dissipation medium being re-evaporated to form the gaseous heat dissipation medium can be shortened, which greatly improves the heat dissipation efficiency of the heat dissipation medium and improves the heat dissipation ability of the heat dissipation device.

(29) In some embodiments, the second spaces **114** can be formed by the two ends of the at least two first capillary structures **120** and the sidewalls of the body **110** in the first direction. The liquid heat dissipation medium in the first capillary structures **120** at the two ends of the at least two first capillary structures **120** can be evaporated to the second spaces **114**. Thus, the liquid heat dissipation medium in the first capillary structures **120** at the two ends of the at least two first capillary structures **120** can be quickly evaporated to form the gaseous heat dissipation medium in the second spaces **114** to flow.

(30) In some embodiments, the two ends of the at least two first capillary structures **120** can refer to two ends of the at least two first capillary structures **120** when being used as a whole.

(31) In some embodiments, the liquid heat dissipation medium in the two neighboring first capillary structures **120** can be evaporated to the first spaces **112**. The liquid heat dissipation medium in the first capillary structures **120** at the two ends of the at least two first capillary structures **120** can be evaporated to the second spaces **114**. Thus, gaseous heat dissipation medium spaces can be formed at two ends of each first capillary structure **120**. Therefore, the liquid heat dissipation medium in the at least two first capillary structures **120** can be quickly evaporated to form the gaseous heat dissipation medium, which greatly improves the heat dissipation ability of the heat dissipation device.

(32) In some embodiments of the present disclosure, as shown in FIG. **4**, a first part of the first capillary structures **120** of the at least two first capillary structures **120** is arranged on the side of the operation wall of the body **110** in parallel in the first direction A. The operation wall **113** can be a sidewall of the body **110** connected to the heat generation element. A second part of the first capillary structures **120** of the at least two first capillary structures **120** can be arranged in parallel in the second direction B. The first direction can satisfy the perpendicular condition with the flow direction of the heat dissipation medium. The second direction and the first direction can satisfy the perpendicular condition. Thus, the cross-section of the single first capillary structure **120** can be set to be relatively small. A first space **112** is formed between two neighboring first capillary structures **120**. Thus, the liquid heat dissipation medium in the single first capillary structure **120** can be evaporated more quickly to form the gaseous heat dissipation medium, which greatly improves the heat dissipation ability of the heat dissipation device.

(33) In some embodiments, a manner of forming the first space **112** by neighboring first capillary structures **120** of the first part of the first capillary structures **120**, and a manner of forming the first

space **112** by neighboring first capillary structures **120** in the second part of the first capillary structures **120** can be similar to the manner of forming the first space **112** between the two neighboring first capillary structures **120** and are not repeated here.

(34) For example, a first interval can be formed between the neighboring first capillary structures **120** in the first part of the first capillary structures **120**. Ends where the neighboring first capillary structures **120** of the second part of the first capillary structures **120** contact with each other can be non-planar.

(35) For example, as shown in FIG. 4, the first interval is formed between the neighboring first capillary structures **120** of the first part of the first capillary structures **120**. The first space **112** can be formed through the first interval. Recesses **121** are arranged on surfaces of a part of the first capillary structures **120** of the second part of the first capillary structures **120** close to the neighboring first capillary structures **120**. The first space **112** can be formed through the recesses **121** of the second part of the first capillary structures **120**.

(36) In some embodiments, the neighboring first capillary structures **120** of the second part of the first capillary structures **120** can contact with each other through a surface-to-surface contact. Thus, the heat of the heat generation element can be transferred between the neighboring first capillary structures **120**. The liquid heat dissipation medium can be used to quickly form the gaseous heat dissipation medium in the recesses **121** through the recesses **121**, which can greatly improve the heat dissipation ability of the heat dissipation device.

(37) In some embodiments, a shape of the cross-section of the recess **121** is not limited. For example, the cross-section of the recess **121** can be semicircular or rectangular.

(38) In some embodiments of the present disclosure, the heat dissipation device can further include a second capillary structure **130**. The second capillary structure **130** can be arranged on the inner surface forming the accommodation chamber **111**. The gaseous heat dissipation medium can be condensed into a liquid state in the second capillary structure **130**. The liquid heat dissipation medium can flow to the first capillary structure **120** based on the second capillary structure **130** and be evaporated again in the first capillary structure **120**.

(39) In some embodiments, at least two first capillary structures **120** can be in contact with the second capillary structure **130**. The heat of the heat generation element can be transferred to the first capillary structures **120** through the second capillary structures **130**. When the heat dissipation device does not include the second capillary structure **130**, the first capillary structure **120** can directly contact the inner surface forming the accommodation chamber **111**.

(40) In the present disclosure, structures of the first capillary structure **120** and the second capillary structure **130** are not limited. For example, the second capillary structure **130** can have a metal mesh structure, and the first capillary structure **120** can have a non-metal structure. Thus, the first capillary structure **120** can have absorbability. The liquid heat dissipation medium can be adsorbed into the first capillary structure **120**. The liquid heat dissipation medium adsorbed into the first capillary structure **120** can absorb the heat and evaporates into the accommodation chamber **111**. The gaseous heat dissipation medium can be condensed in the second capillary structure **130**.

(41) The heat dissipation device of embodiments of the present disclosure can include the body **110** having the accommodation chamber **111**, the heat dissipation medium arranged in the accommodation chamber **111**, and the at least two first capillary structures **120**. The at least two first capillary structures **120** can be in a stripe shape and can be arranged in the accommodation chamber **111** along the flow direction of the heat dissipation medium. The first space **112** can be formed between the two neighboring first capillary structures **120**. The liquid heat dissipation medium in the two neighboring first capillary structures **120** can be evaporated into the first space **112**. A flow channel of the gaseous heat dissipation medium can be formed by the first space **112**. The liquid heat dissipation medium in the two neighboring first capillary structures **120** can be used to quickly form the gaseous heat dissipation medium into the first space **112** to flow. Thus, the route and time of forming the gaseous heat dissipation medium from the liquid heat dissipation

medium can be greatly reduced, and the heat dissipation ability of the heat dissipation device can be improved.

(42) Embodiments of the present disclosure further provide an electronic apparatus. The electronic apparatus can include the heat dissipation device and heat generation element of embodiments of the present disclosure. The heat dissipation device can be configured to dissipate heat for the heat generation element.

(43) In some embodiments, the structure of the electronic apparatus is not limited. For example, the electronic apparatus can include a computer or a mobile phone.

(44) The above description only describes embodiments of the present disclosure. However, the scope of the present disclosure is not limited here. Those skilled in the art can easily think of the changes or replacements within the technical scope of the present disclosure. These changes and replacements shall be within the scope of the present disclosure. Therefore, the protection scope of the present invention shall be subjected to the scope of the claims.

Claims

1. A heat dissipation device, comprising: a body including an accommodation chamber, the accommodation chamber including a heat dissipation medium; at least two first capillary structures being stripe-shaped and arranged in the accommodation chamber along a flow direction of the heat dissipation medium, a first space being formed between two neighboring first capillary structures, wherein: a liquid heat dissipation medium in the two neighboring first capillary structures is evaporated into the first space; a first capillary structure of the at least two first capillary structures includes: a first stripe-shaped member having a cross-section; and a second stripe-shaped member having a cross-section different from the cross-section of the first stripe-shaped member and being alternatively arranged with the first stripe-shaped member to form the first capillary structure.
2. The heat dissipation device according to claim 1, wherein: the at least two first capillary structures are arranged in parallel in a first direction on a side of an operation wall of the body; the operation wall is a wall on a side where the body is connected to the heat generation element; and the first direction and the flow direction of the heat dissipation medium satisfy a perpendicular condition.
3. The heat dissipation device according to claim 2, wherein: second spaces are formed between two ends of the at least two first capillary structures and a sidewall of the body in the first direction; and liquid heat dissipation medium of the first capillary structures at the two ends of the at least two first capillary structures is evaporated into the second space.
4. The heat dissipation device according to claim 3, wherein: the two neighboring first capillary structures are in contact; or a first interval is formed between the two neighboring first capillary structures.
5. The heat dissipation device according to claim 1, wherein the first capillary structure further includes: a third stripe-shaped member being staggered with the first stripe-shaped member and the second stripe-shaped member to form the first capillary structure; wherein: the third stripe-shaped member, the first stripe-shaped member, and the second stripe-shaped member are column-shaped; and at least two of the first stripe-shaped member, the second stripe-shaped member, and the third stripe-shaped member have different diameters.
6. The heat dissipation device according to claim 1, wherein: a first part of first capillary structures of the at least two first capillary structures are arranged in parallel in a first direction on a side of an operation wall of the body, and the operation wall is a wall on a side where the body is connected to a heat generation element; a second part of first capillary structures of the at least two first capillary structures are arranged in parallel in a second direction; and the first direction and the flow direction of the heat dissipation medium satisfy a perpendicular condition, and the second direction and the first direction satisfy the perpendicular condition.

7. The heat dissipation device according to claim 1, wherein thicknesses of the at least two first capillary structures in an evaporation region are greater than thicknesses of the at least two first capillary structures in a condensation region.
8. The heat dissipation device according to claim 1, further comprising: a second capillary structure arranged on the inner surface of the accommodation chamber, a gaseous heat dissipation medium being condensed into a liquid state in the second capillary structure; wherein the second capillary structure has a metal mesh structure, and the first capillary structure has a non-metal structure.
9. An electronic apparatus comprising: a heat dissipation device including: a body including an accommodation chamber, the accommodation chamber including a heat dissipation medium; and at least two first capillary structures being stripe-shaped and arranged in the accommodation chamber along a flow direction of the heat dissipation medium, a first space being formed between two neighboring first capillary structures, wherein: a liquid heat dissipation medium in the two neighboring first capillary structures is evaporated into the first space; a first capillary structure of the at least two first capillary structures includes: a first stripe-shaped member having a cross-section; and a second stripe-shaped member having a cross-section different from the cross-section of the first stripe-shaped member and being alternatively arranged with the first stripe-shaped member to form the first capillary structure; and a heat generation element, the heat dissipation device being configured to dissipate heat of the heat generation element.
10. The electronic apparatus according to claim 9, wherein: the at least two first capillary structures are arranged in parallel in a first direction on a side of an operation wall of the body; the operation wall is a wall on a side where the body is connected to the heat generation element; and the first direction and the flow direction of the heat dissipation medium satisfy a perpendicular condition.
11. The electronic apparatus according to claim 10, wherein: second spaces are formed between two ends of the at least two first capillary structures and a sidewall of the body in the first direction; and liquid heat dissipation medium in first capillary structures at the two ends of the at least two first capillary structures is evaporated into the second space.
12. The electronic apparatus according to claim 11, wherein: the two neighboring first capillary structures are in contact; or a first interval is formed between the two neighboring first capillary structures.
13. The electronic apparatus according to claim 9, wherein the first capillary structure further includes: a third stripe-shaped member being staggered with the first stripe-shaped member and the second stripe-shaped member to form the first capillary structure; wherein: the third stripe-shaped member, the first stripe-shaped member, and the second stripe-shaped member are column-shaped; and at least two of the first stripe-shaped member, the second stripe-shaped member, and the third stripe-shaped member have different diameters.
14. The electronic apparatus according to claim 9, wherein: a first part of first capillary structures of the at least two first capillary structures are arranged in parallel in a first direction on a side of an operation wall of the body, and the operation wall is a wall on a side where the body is connected to a heat generation element; a second part of first capillary structures of the at least two first capillary structures are arranged in parallel in a second direction; and the first direction and the flow direction of the heat dissipation medium satisfy a perpendicular condition, and the second direction and the first direction satisfy the perpendicular condition.
15. The electronic apparatus according to claim 9, wherein thicknesses of the at least two first capillary structures in an evaporation region are greater than thicknesses of the at least two first capillary structures in a condensation region.
16. The electronic apparatus according to claim 9, wherein the heat dissipation device further includes: a second capillary structure arranged on the inner surface of the accommodation chamber, a gaseous heat dissipation medium being condensed into a liquid state in the second capillary structure; wherein the second capillary structure has a metal mesh structure, and the first capillary structure has a non-metal structure.

